

CLARE GATHOGA

LECTURE THREE: PARTIAL DIFFERENTIATION CONCEPT OF TOTAL DIFFERENTIAL

A function of two or more independent variables is called a multivariate function. The total change of the dependent variable will be as a result of the changes resulting from a change of each of the independent variable.

For example, when we consider the change of utility of a household which consumes two types of food, ugali and rice, we can represent it as:

$$U = f(X_1, X_2)$$

Where U = total utility derived

X_1 = units of ugali consumed

X_2 = units of rice consumed

When analyzing the total change in the utility of the household, due to change in number of units of each food item consumed, it will be composed of two parts.

- i) Change in utility due to change in units of ugali consumed
- ii) Change in utility due to change in units of rice consumed

Symbolically: the total change in utility is given as:

$$dU = \frac{\partial U}{\partial X_1} dX_1 + \frac{\partial U}{\partial X_2} dX_2$$

The sum of the two parts is called the total differential of the utility function. i.e

dU = Total differential of utility. It is the total change in utility as a result of change in number of units of ugali and number of units in rice consumed.

Each of the parts is called a partial differential of the utility function with respect to goods 1 and 2 respectively.

$$\frac{\partial U}{\partial X_1} dX_1 = \text{Partial differential of utility function with respect to } X_1$$

. It measures the change in utility due to amount of good 1(Ugali) consumed

$\frac{\partial U}{\partial X_2} dX_2$ = Partial differential of utility function with respect to X_2 . It measures the change in utility due to the amount of good 2 (rice) consumed.

In general, if a function is given as:

$$Y = f(X_1, X_2, X_3, \dots \dots \dots X_n)$$

The total differential is given as:

$$dY = \frac{\partial Y}{\partial X_1} dX_1 + \frac{\partial Y}{\partial X_2} dX_2 + \frac{\partial Y}{\partial X_3} dX_3 \dots \dots \dots + \frac{\partial Y}{\partial X_n} dX_n$$

Where $\frac{\partial Y}{\partial X_i} dX_i$ represents the change in dependent variable Y due to change in variable i

If we are interested in knowing how Y changes when X_i changes, the other variables are assumed to be constant.

Example 1

$$Y = 4X_1 + 10X_2 X_1^2 + 3X_2^3$$

Find the total differential of the function

$$dY = \frac{\partial Y}{\partial X_1} dX_1 + \frac{\partial Y}{\partial X_2} dX_2$$

$\frac{\partial Y}{\partial X_1}$ = Partial derivative of Y with respect to X_1 when X_2 is held constant.

$\frac{\partial Y}{\partial X_2}$ = Partial derivative of Y with respect to X_2 when X_1 is held constant.

$$dY = (4 + 20X_2X_1) dX_1 + (10X_1^2 + 9X_2^2)dX_2$$

Example 2

Get the total differential of the function below:

$$Z = 5XY + 2X^2Y - 8Y^2X$$

Solution

$$dZ = (5Y + 4XY - 8Y^2)dX + (5X + 2X^2 - 16YX)dY$$

3.4 Partial differentiation:

This is applied to multivariable functions when we are analysing the change in the dependent variables as a result of change in one of the independent variables when the other independent variables are held constant.

For example, A firm produces 2 goods 1 and 2. If we want to know the change in the total cost of the firm when the production of good 1 is increased when production of good 2 is held constant, partial derivatives of the cost function with respect to change in good 1 will be useful.

Process of getting the partial derivative:

If a dependent variable Y is a function of two independent variables X_1 and X_2 , symbolically.

$$Y = f(X_1, X_2)$$

Suppose X_1 changes by an amount ΔX_1 when X_2 is held constant, Y will change by an amount ΔY . this will be given as:

$$Y + \Delta Y = f(X_1 + \Delta X_1, X_2)$$

$$\Delta Y = f(X_1 + \Delta X_1, X_2) - Y$$

$$\text{Since } Y = f(X_1, X_2)$$

$$\Delta Y = f(X_1 + \Delta X_1, X_2) - f(X_1, X_2)$$

The average change in y with respect to x_1 will be given as:

$$\frac{\Delta Y}{\Delta X_1} = \frac{f(X_1 + \Delta X_1, X_2) - f(X_1, X_2)}{\Delta X_1}$$

The partial derivative of Y with respect to X_1 is the limit of: $\frac{\Delta Y}{\Delta X_1}$ as ΔX_1 approaches 0

This is represented as:

$$\frac{\partial Y}{\partial X_1} = \frac{f(X_1 + \Delta X_1, X_2) - f(X_1, X_2)}{\Delta X_1}$$

On the other hand, assume X_2 changes when X_1 is held constant. Y will change and will be given as:

$$Y + \Delta Y = f(X_1, X_2 + \Delta X_2)$$

$$\Delta Y = f(X_1, X_2 + \Delta X_2) - f(X_1, X_2)$$

$$\frac{\Delta Y}{\Delta X_2} = \frac{f(X_1, X_2 + \Delta X_2) - f(X_1, X_2)}{\Delta X_2}$$

The partial derivative as $\Delta x_2 \rightarrow 0$ is given as:

$$\frac{\Delta Y}{\Delta X_2} = \lim_{\Delta X_2 \rightarrow 0} \frac{f(X_1, X_2 + \Delta X_2) - f(X_1, X_2)}{\Delta X_2} = \frac{\partial Y}{\partial X_2}$$

$$\frac{\partial Y}{\partial X_1} \text{ can be represented as } f_{x_1} \text{ or } f_1$$

$$\text{While } \frac{\partial Y}{\partial X_2} \text{ can be represented as } f_{x_2} \text{ or } f_2$$

Example 3

$$Y = 5X_1 + 2X_2$$

$$\text{find } \frac{\partial Y}{\partial X_1} \text{ and } \frac{\partial Y}{\partial X_2}$$

$$Y = 5X_1 + 2X_2$$

Suppose X_1 changes by ΔX_1 and X_2 is held constant. Now Y will be given as:

$$Y + \Delta Y = 5(X_1 + \Delta X_1) + 2X_2$$

$$Y + \Delta Y = 5(X_1 + \Delta X_1) + 2X_2 - Y$$

$$\Delta Y = \cancel{5X_1} + 5\Delta X_1 + \cancel{2X_2} - \cancel{5X_1} - \cancel{2X_2}$$

$$\Delta Y = 5\Delta X_1$$

$$\frac{\Delta y}{\Delta x_1} = 5$$

$$\frac{\partial y}{\partial x_1} = \lim_{\Delta x_1 \rightarrow 0} \frac{\Delta y}{\Delta x_1} = 5$$

On the other hand, if X_2 changes by ΔX_2 when X_1 is held constant, new Y will be:

$$Y + \Delta Y = 5X_1 + 2(X_2 + \Delta X_2)$$

$$\Delta Y = \cancel{5X_1} + \cancel{2X_2} + 2\Delta X_2 - \cancel{5X_1} - \cancel{2X_2}$$

$$\frac{\Delta y}{\Delta x_2} = 2$$

$$\frac{\partial y}{\partial x_2} = \lim_{\Delta x_2 \rightarrow 0} \frac{\Delta y}{\Delta x_2} = 2$$

Example 4

$$Y = 4X_2^2 + 4X_1 X_2 + 5X_1^2$$

Get $\frac{\partial y}{\partial x_1}$ and $\frac{\partial y}{\partial x_2}$

1. Change in Y when X_1 changes

$$(Y + \Delta Y) = 4X_2^2 + 4X_2(X_1 + \Delta X_1) + 5(X_1 + \Delta X_1)^2$$

$$\Delta Y = 4X_2^2 + 4X_2X_1 + 4X_2\Delta X_1 + 5X_1^2 + 10X_1\Delta X_1 + 5\Delta X_1^2 - (4X_2^2 + 4X_1X_2 + 5X_1^2)$$

$$= 4X_2\Delta X_1 + 10X_1\Delta X_1 + 5\Delta X_1^2$$

$$\frac{\Delta Y}{\Delta X_1} = 4X_2 + 10X_1 + 5\Delta X_1$$

Using the limits of the function as ΔX_1 approaches 0, we will get the partial derivative of Y with respect to X_1

$$\frac{\partial Y}{\partial X_1} = 4X_2 + 10X_1$$

2. Change in Y when X₂ changes

$$Y + \Delta Y = 4(X_2 + \Delta X_2)^2 + 2X_1(X_2 + \Delta X_2) + 5X_1^2$$

$$\Delta Y = 4X_2^2 + 8X_2 \Delta X_2 + \Delta X_2^2 + 2X_1 X_2 + 2X_1 \Delta X_2 + 5X_1^2 - 4X_2^2 - 2X_1 X_2 - 5X_1^2$$

$$\frac{\Delta Y}{\Delta X_2} = 8X_2 \Delta X_2 + \Delta X_2 + 2X_1 \Delta X_2 / \Delta X_2 = 8X_2 + \Delta X_2 + 2X_1$$

Using the limits of the function as ΔX_2 approaches 0, we will get the partial derivative of Y with respect to X_2

$$\frac{\partial Y}{\partial X_2} = 8X_2 + 2X_1$$

3.5 Rules of partial differentiation

Whenever we differentiate multivariate function with respect to one of the independent variable, the other variables are treated as constants.

Example

$$Y = 15X_1 + 8X_2^2$$

$$\frac{\partial Y}{\partial X_1} = 15 \quad (8x_2^2 \text{ behaves like a constant hence its derivative with respect to } X_1 \text{ is } 0)$$

$$\frac{\partial Y}{\partial X_2} = 16X_2 \quad (15x_1 \text{ is a constant. Its derivative with respect to } X_2 \text{ is } 0)$$

If:

$$Y = 2X_1 X_2 + 3X_1^2 + 12X_2^2$$

$$\frac{\partial Y}{\partial X_1} = 2X_2 + 6X_1 \quad (12X_2^2 \text{ behaves as constant hence its derivative with respect to } X_1 \text{ is } 0, \text{ but in } 2X_1 X_2, X_2 \text{ is in a multiplicative function, hence its retained as a constant})$$

$$\frac{\partial Y}{\partial X_2} = 2X_1 + 24X_2$$

1. Sum and difference rule.

$$\text{If } Y = f(X_1, X_2) \pm g(X_1, X_2)$$

Then:

$$\frac{\partial Y}{\partial X_1} = f'(X_1) + g'(X_1)$$

Similarly

$$\frac{\partial Y}{\partial X_2} = f'(X_2) + g'(X_2)$$

Example

Get $\frac{\partial y}{\partial x_1}$ and $\frac{\partial y}{\partial x_2}$ in the following function

i. $Y = 8X_1 + 5X_1^2 X_2$

$$\frac{\partial Y}{\partial X_1} = 8 + 10X_1 X_2$$

$$\frac{\partial Y}{\partial X_2} = 5X_1^2$$

ii. $Y = 15X_2^2 X_1^3 - 2X_1^2 X_2$

$$\frac{\partial Y}{\partial X_1} = 45X_2^2 X_1^2 - 4X_1 X_2$$

$$\frac{\partial Y}{\partial X_2} = 30X_2 X_1^3 - 2X_1^2$$

Practice question

$$Y = 5X_1^2 + 3X_1 X_2 - 4X_1^3 X_2^2$$

Get $\frac{\partial y}{\partial x_1}$ and $\frac{\partial y}{\partial x_2}$

2. Product rule

If we have a function Y given as:

$$Y = f(X_1, X_2) \cdot g(X_1, X_2)$$

The partial derivatives will be given as:

$$\frac{\partial Y}{\partial X_1} = \frac{df}{dX_1} \cdot g(X_1, X_2) + \frac{dg}{dX_1} f(X_1, X_2)$$

Similarly

$$\frac{\partial Y}{\partial X_2} = \frac{df}{dX_2} \cdot g(X_1, X_2) + \frac{dg}{dX_2} f(X_1, X_2)$$

Example:

$$Y = (8X_1^2 + 2X_2) \cdot (4X_2^3 + 3X_1 X_2)$$

$$\frac{\partial Y}{\partial X_1} = 16X_1(4X_2^3 + 3X_1 X_2) + 3X_2(8X_1^2 + 2X_2)$$

$$= 64X_1 X_2^3 + 48X_1^2 X_2 + 24X_2 X_1^2 + 6X_2^2$$

$$= 64X_1 X_2^3 + 72X_1^2 X_2 + 6X_2^2$$

$$\begin{aligned}
\frac{\partial Y}{\partial X_2} &= 2(4X_2^3 + 3X_1X_2) + (12X_2^2 + 3X_1)(8X_1^2 + 2X_2) \\
&= 8X_2^3 + 6X_1X_2 + 96X_2^2X_1^2 + 24X_2^3 + 24X_1^3 + 6X_1X_2 \\
&= 32X_2^3 + 12X_1X_2 + 96X_2^2X_1^2 + 24X_1^3
\end{aligned}$$

3. Quotient rule

$$Y = \frac{f(X_1, X_2)}{g(X_1, X_2)}$$

$$\frac{\partial Y}{\partial X_1} = \frac{\frac{df}{dX_1} \cdot g(X_1, X_2) - \frac{dg}{dX_1} f(X_1, X_2)}{(g(X_1, X_2))^2}$$

$$\frac{\partial Y}{\partial X_2} = \frac{\frac{df}{dX_2} \cdot g(X_1, X_2) - \frac{dg}{dX_2} f(X_1, X_2)}{(g(X_1, X_2))^2}$$

Example:

$$Y = \frac{8X_1^2X_2}{4X_1 + 3X_2^2}$$

$$f(X_1) = 64X_1^2X_2 \quad g(X_1) = 4$$

$$f(X_2) = 8X_1^2 \quad g(X_2) = 6X_2$$

$$\frac{\partial Y}{\partial X_1} = \frac{16X_1X_2 \cdot (4X_1 + 3X_2^2) - 4(8X_1^2 \cdot X_2)}{(4X_1 + 3X_2^2)^2}$$

$$\frac{\partial Y}{\partial X_1} = \frac{64X_1^2X_2 + 48X_1X_2^3 - 32X_1^2 \cdot X_2}{(4X_1 + 3X_2^2)^2}$$

$$\frac{\partial Y}{\partial X_1} = \frac{32X_1^2X_2 + 48X_1X_2^3}{(4X_1 + 3X_2^2)^2}$$

$$\frac{\partial Y}{\partial X_2} = \frac{8X_1^2 \cdot (4X_1 + 3X_2^2) - 6X_2(8X_1^2 \cdot X_2)}{(4X_1 + 3X_2^2)^2}$$

$$\frac{\partial Y}{\partial X_2} = \frac{32X_1^3 - 24X_1^2X_2^2}{(4X_1 + 3X_2^2)^2}$$

4. Chain rule - derivative of a function of 2 two independent variables raised to power

$$Y = [f(X_1, X_2)]^n$$

$$\frac{\partial Y}{\partial X_1} = n. [f(X_1, X_2)]^{n-1} \cdot f'_{X_1}$$

$$\frac{\partial Y}{\partial X_2} = n. [f(X_1, X_2)]^{n-1} \cdot f'_{X_2}$$

Example

$$Y = (8X_2^2 + 3X_1 X_2)^4$$

$$\frac{\partial Y}{\partial X_1} = 4. [8X_2^2 + 3X_1 X_2]^3 \cdot (3X_2)$$

$$\frac{\partial Y}{\partial X_1} = 12X_2 \cdot [8X_2^2 + 3X_1 X_2]^3$$

$$\frac{\partial Y}{\partial X_2} = 4. [8X_2^2 + 3X_1 X_2]^3 \cdot (16X_2 + 3X_1)$$

$$\frac{\partial Y}{\partial X_2} = (48X_2 + 12X_1) \cdot [8X_2^2 + 3X_1 X_2]^3$$

3.6 First order and second order partial derivative

First order partial derivative

If we have a multivariate function given as:

$$Y = f(X_1, X_2)$$

Then:

$\frac{\partial Y}{\partial X_1}$ and $\frac{\partial Y}{\partial X_2}$ are called the first order partial derivatives with respect to X_1 and X_2 respectively.

When the first order partial derivatives are differentiated again with respect to X_1 and X_2 we get the second order partial derivatives and the cross partial derivatives.

ii) Second order partial derivatives:

$$\text{Given : } Y = f(X_1, X_2)$$

Then:

$\frac{\partial Y}{\partial X_1}$ can be written as: $\frac{\partial Y}{\partial X_1} = f_{X_1}$ while $\frac{\partial Y}{\partial X_2} = f_{X_2}$

The second order partial derivative of Y with respect to X_1 is given as:

$$\frac{\partial^2 Y}{\partial X_1^2} = f_{X_1 X_1}$$

While the second order partial derivative of Y with respect to X_2 is given as:

$$\frac{\partial^2 Y}{\partial X_2^2} = f_{X_2 X_2}$$

In addition we get the partial derivatives of $\frac{\partial Y}{\partial X_1}$ with respect to X_2 and $\frac{\partial Y}{\partial X_2}$ with respect to X_1 . These are given as:

$$\frac{\partial^2 Y}{\partial X_1 \partial X_2} = f_{X_1 X_2} \text{ This tells us how } f_{X_1} \text{ changes when } X_2 \text{ changes.}$$

And

$$\frac{\partial^2 Y}{\partial X_2 \partial X_1} = f_{X_2 X_1} \text{ tells us how } f_{X_2} \text{ changes when } X_1 \text{ changes.}$$

These two second order partial derivatives are known as the cross partial derivatives.

According to the young's theorem, the cross partial derivatives should be equal. i.e.

$$f_{X_1 X_2} = f_{X_2 X_1}$$

Example:

Given a function

$$Y = 3X_1^2 + 2X_2^3 + 6X_1 X_2$$

Get the first and the second order partial derivatives with respect to X_1 and X_2 .

$$\frac{\partial Y}{\partial X_1} = 6X_1 + 6X_2 \text{ while } \frac{\partial Y}{\partial X_2} = 6X_2^2 + 6X_1$$

$$\frac{\partial^2 Y}{\partial X_1^2} = f_{X_1 X_1} = 6$$

$$\frac{\partial^2 Y}{\partial X_2^2} = f_{X_2 X_2} = 12X_2$$

$$\frac{\partial^2 Y}{\partial X_1 \partial X_2} = f_{X_1 X_2} = 6$$

$$\text{and } \frac{\partial^2 Y}{\partial X_2 \partial X_1} = f_{X_2 X_1} = 6$$

The young's theorem is proved:

$$f_{X_1 X_2} = f_{X_2 X_1} = 6$$

Assignment

Get the partial derivatives of the following functions:

i) $Y = f(X_1, X_2) = [18X_1^2 + 3X_2 X_1 - X_2^2]^4$

ii) $y = \frac{8X_1^2}{(2X_1 + 3X_2 X_1)^3}$

$$\text{iii) } Y = (4X_1 + 8X_2^2)^2 (9X_2^2 + 2X_1 X_2)$$

Get the first order, second order and cross partial derivatives of the following functions.

$$\text{i) } Y = 8X_1 + 15X_1^2 X_2 + 3X_2^3$$

$$\text{ii) } Y = 2X^2 + 3XW + 7W^3$$

LECTURE FOUR: ECONOMIC APPLICATIONS OF PARTIAL DERIVATIVES

Multivariate functions in economics

In Economics, most functions are multivariate. We therefore need to look at how these functions change due to change in one of the independent variable when others are held constant.

4.4 Utility function and the marginal utility

Assume a consumer derives satisfaction from consuming 2 goods 1 and 2. The utility function can be given as

$$U = f(Q_1, Q_2)$$

Where U = Total utility

Q_1 = Amount of good 1 consumed

Q_2 = Amount of good 2 consumed.

We assume that this utility function is continuous, differentiable and 2nd order partial derivatives can be obtained.

If each of the 2 goods has positive marginal utility, then

$$\frac{\partial U}{\partial Q_1} > 0 \quad \frac{\partial U}{\partial Q_2} > 0$$

$$\frac{\partial U}{\partial Q_1} = \text{marginal utility of good 1}$$

$$\frac{\partial U}{\partial Q_2} = \text{marginal utility of good 2}$$

Total differential of the utility function

$$dU = \frac{\partial U}{\partial X_1} dX_1 + \frac{\partial U}{\partial X_2} dX_2$$

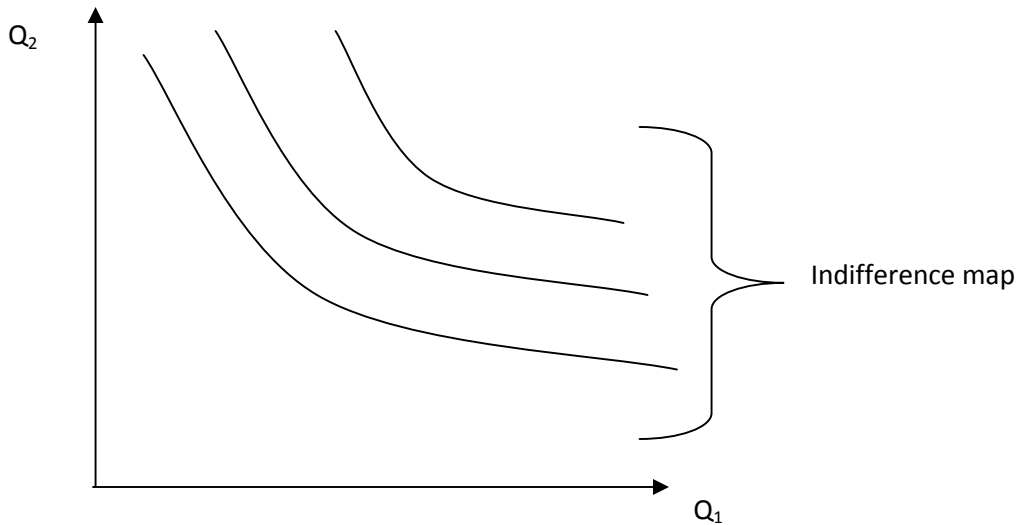
$$dU = MU_{X_1} dX_1 + MU_{X_2} dX_2$$

If you recall what you learnt in micro economics I, an indifference curve is convex to the origin and its downward sloping. Along an indifference curve, the utility level is constant and does not change. Hence, $dU = 0$

$$0 = MU_{X_1} dX_1 + MU_{X_2} dX_2$$

The slope of an indifference curve is given as:

$$\frac{dX_2}{dX_1} = \frac{-MU_{X_1}}{MU_{X_2}}$$



If the slope of the marginal utility with respect to a good is positive, the utility function displays increasing marginal utility with respect to that good and if negative, it displays diminishing marginal utility. Most normal goods display decreasing marginal utility hence;

$\frac{\partial MU(Q_1)}{\partial Q_1}$ is expected to be less than 0.

$$\frac{\partial MU(Q_1)}{\partial Q_1} = \frac{\partial^2 U}{\partial Q_1^2}$$

Example one

A utility function is given as:

$$U = Q_1^{1/2} Q_2^{4/3}$$

- i) Determine the marginal utility of good 1 and 2
- ii) Determine the nature of marginal utility for each of the goods.

Solution

- i) $\frac{\partial U}{\partial Q_1} = 0.5 Q_1^{-0.5} Q_2^{4/3} =$ marginal utility of good 1
 $\frac{\partial U}{\partial Q_2} = \frac{4}{3} Q_1^{0.5} Q_2^{1/3}$ marginal utility of good 2
- ii) $\frac{\partial (MU_{Q_1})}{\partial Q_1} = -0.25 Q_1^{-1.5} Q_2^{4/3} = \frac{-0.25 Q_2^{4/3}}{Q_1^{1.5}}$ Given that Q_1 and Q_2 are non-negative, it means the entire expression will be negative, hence good 1 displays decreasing marginal utility.
- iii) $\frac{\partial (MU_{Q_2})}{\partial Q_2} = \frac{4}{9} Q_1^{0.5} Q_2^{-2/3} = \frac{4}{9} \frac{Q_1^{0.5}}{Q_2^{2/3}}$ Given that Q_1 and Q_2 are non-negative, it means the entire expression will be positive, hence good 2 displays increasing marginal utility.

4.5 COST FUNCTION

We have been assuming that a firm produces 1 product only. But in some cases, a firm is likely to produce several products. There is need to find out how the total cost of a firm changes when the production one of the products change when the production of other products is held constant. The total cost of the firm will be a function of production level of each of the products. If we have n products then

$$TC = f(Q_1, Q_2, \dots, Q_n)$$

Assuming we have 2 products only then

$$TC = f(Q_1, Q_2)$$

The marginal costs of good 1 is given as:

and 2 are

$$MC_1 = \frac{dTC}{dQ_1} \quad \text{While marginal cost of good 2 is given as: } MC_2 = \frac{dTC}{dQ_2}$$

Example two

$$C = 8Q_1^3 + 6Q_2^2 + 4Q_1Q_2 - 10$$

$$MC_1 = \frac{dTC}{dQ_1} = 24Q_1^2 + 4Q_2 \quad MC_2 = \frac{dTC}{dQ_2} = 12Q_2 + 4Q_1$$

The firm is also interested in knowing if marginal costs are increasing or decreasing with output change. This is computed by getting the derivatives of partial marginal cost functions with respect to that good.

In our previous example:

$$MC_1 = \frac{dTC}{dQ_1} = 24Q_1^2 + 4Q_2$$

To determine if MC_1 increases or decreases with output we differentiate it with respect to Q_1

$\frac{dMC_1}{dQ_1} = 48Q_1 > 0$ since Q_1 is positive, the whole expression will be positive, hence good 1 displays increasing marginal cost.

Similarly, To determine if MC_2 increases or decreases with output we differentiate it with respect to Q_2 .

$$\frac{dMC_2}{dQ_2} = 12 > 0, \text{ hence good 2 displays increasing marginal cost.}$$

4.6 PARTIAL ELASTICITY OF DEMAND

In partial market equilibrium, the demand for a product is assumed to be a function of its own price when other factors are held constant.

$$Qdx = f(Px) \text{ ceteris paribus}$$

$$Qdy = f(Py)$$

This is a very simplified market model. In most cases, the demand of a product will be affected by its own price, and also the prices of other related commodities such as its substitutes and complements, the income among other variables.

Assuming we have a product A, its demand function can be given as:

$$Q_A = f(P_A, P_B, Y)$$

Where:

Q_A = quantity demanded of good A

P_A = Price of good A

P_B = Price of good B

Y = consumer income

We will now look at how we compute the responsiveness of the quantity demanded of good A with respect to price of good A, price of good B and the income.

Price elasticity of demand

It measures the percentage change in the quantity demanded of good A when its own price changes keeping the price of good B (P_B) and the income (Y) constant.

$$\text{Price elasticity of demand} = \frac{\partial Q_A \cdot P_A}{\partial P_A \cdot Q_A}$$

Cross price elasticity of demand

It measures the percentage change in quantity demanded of good A when price of good B changes when the price of good A (P_A) and other factors such as the income (Y) are held constant.

$$\text{Cross price elasticity of demand} = \frac{\partial Q_A \cdot P_B}{\partial P_B \cdot Q_A}$$

You can know whether two goods are substitutes or compliments by looking at the sign of the partial derivatives of their demand functions with respect to price of the other good e.g.

$\frac{\partial Q_A}{\partial P_B} > 0$ the two goods are substitutes i.e. As price of good B increase, the demand of good A increase. If demand for a good goes down due to an increase in its own price and consumers substitute it by buying more its substitute.

For complementary commodities

$$\frac{\partial Q_A}{\partial P_B} < 0$$

As price of good B increase quantity of demanded of good A decrease. The decrease in demand for good B due to an increase in its price causes the demand of good A to decrease since the two goods are used together.

Income elasticity of demand

It measures the percentage change in the quantity demanded of good A when income changes when the price of good A and the price of good B are held constant.

$$\text{Income elasticity of demand} = \frac{\partial Q_A \cdot Y}{\partial Y \cdot Q_A}$$

The income elasticity is positive for normal goods and negative for inferior and giffen goods.

Example Three

A demand function is given as:

$$Q_A = 90 - 2P_A + 0.4P_B + 0.1Y$$

Find the price, income and cross price elasticities of demand when:

$$P_A = 6 \quad P_B = 10 \quad Y = 1000$$

$$Q_A = 90 - 2(6) + 0.4(10) + 0.1(1000) = 90 - 12 + 4 + 100 = 182$$

Price elasticity of demand

$$\frac{\partial Q_A}{\partial P_A} = -2$$

$$= \frac{\partial Q_A \cdot P_A}{\partial P_A \cdot Q_A} = -2 * \frac{6}{182} = -0.13$$

A 1% change in the price of good A, holding P_B and Y constant leads to a 0.13% decrease in the quantity demanded of good A. It is inelastic to price change.

Cross-Price elasticity of demand

$$\frac{\partial Q_A}{\partial P_B} = 0.4$$

$$= \frac{\partial Q_A \cdot P_B}{\partial P_B \cdot Q_A} = 0.4 * \frac{10}{92} = 0.04$$

A 1% change in the price of good B, holding P_A and Y constant leads to a 0.04% increase in the quantity demanded of good A. It is inelastic to price change. Since the cross elasticity is positive the two goods are substitutes.

Income elasticity of demand

$$\begin{aligned} \frac{\partial Q_A}{\partial Y} &= 0.1 \\ &= \frac{\partial Q_A \cdot Y}{\partial Y \cdot Q_A} = 0.1 * \frac{100}{92} = 0.11 \end{aligned}$$

A 1% change in the income holding P_B and P_A constant leads to a 0.11% increase in the quantity demanded of good A. It is inelastic to income change. Since the income elasticity is positive, good A is a normal good.

Practice question

Determine whether these 2 goods are substitutes or complements.

$$Qdx = 20 - 3Px + 4Py + 0.2Y$$

$$Px = 1 \quad Py = 2 \quad Y = 500$$

4.7 Production function

In mathematics for economics 1, you learnt about the production function. The production function describes the output level as a function of the level of inputs employed by the firm. If we assume we have only two factors of production, capital and labour, we can denote the production function as:

$$Q = f(L, K)$$

If we want to know how the level of output changes when one of the inputs is changed while the other one is held constant, we use the partial derivatives. The marginal product of labour is the partial derivative of the production function with respect to labour when capital is held constant.

i.e.

$$MP_L = \frac{\partial Q}{\partial L}$$

On the other hand the marginal product of capital is the partial derivative of the production function with respect to capital when the labour is held constant

$$MP_K = \frac{\partial Q}{\partial K}$$

Slope of an isoquant

It we get the total differential of a production function, it will be given as:

$$dQ = \frac{\partial Q}{\partial L} dL + \frac{\partial Q}{\partial K} dK$$

Along an isoquant the output level does not change, hence the total differential, $dQ=0$

If we equate the total differential to 0, we get:

$$dQ = \frac{\partial Q}{\partial L} dL + \frac{\partial Q}{\partial K} dK = 0$$

$$MP_L dL + MP_K dK = 0$$

$$MP_K dK = -MP_L dL$$

Assuming that K is on the Y axis and L is on the X axis, the slope will be given as:

$$\frac{dK}{dL} = -\frac{MP_L}{MP_K}$$

This is the marginal rate of technical substitution (MRTS).

If the derivative of the marginal product of labour with respect to labour is positive, the marginal product is said to be increasing as labour use increases. On the other hand if it is negative, there is diminishing marginal product. If it is 0, it has constant marginal product.

Example Four

A production function is given as:

$$Q = 80K^{0.2}L^{0.5}$$

Determine the marginal product of labour, marginal product of capital, the slope of the production function and whether the marginal products are increasing or decreasing.

Solution

$$MP_L = \frac{\partial Q}{\partial L} = 40K^{0.2}L^{-0.5} = \frac{40K^{0.2}}{L^{0.5}}$$

$$MP_K = \frac{\partial Q}{\partial K} = 16K^{-0.8}L^{0.5} = \frac{16L^{0.5}}{K^{0.8}}$$

$$Slope = \frac{dK}{dL} = -\frac{MP_L}{MP_K} = -\frac{40K^{0.2}}{L^{0.5}} \div \frac{16L^{0.5}}{K^{0.8}} = -\frac{5K}{L}$$

$$\frac{\partial MP_L}{\partial L} = \frac{\partial^2 Q}{\partial L^2} = -20K^{0.2}L^{-1.5} = \frac{-20K^{0.2}}{L^{1.5}} < 0$$

Since K and L are positive, the partial derivative of MP_L with respect to L will be negative. Hence marginal product of labour decreases as more labour is employed.

$$\frac{\partial MP_K}{\partial K} = \frac{\partial^2 Q}{\partial K^2} = -6.4K^{-1.8}L^{0.5} = \frac{-6.4L^{0.5}}{K^{1.8}} < 0$$

There is diminishing marginal product of capital as capital use increases.

4.8 Homogeneity of a function

Given a function $Y = f(X_1, X_2)$

We say it is homogenous of degree r if by multiplying all the independent variables by a constant K, the dependent variable (Y) changes by a multiple K^r

Example five:

$$Y = f(X_1, X_2) = 4X_1 + X_2$$

New $Y = 4kX_1 + kX_2 = k(4X_1 + X_2) = k^1Y$ The function is homogenous of degree 1. Functions which are homogenous of degree 1 are said to be linearly homogenous.

Determine the homogeneity of the function given as:

$$Y = X^2 + 5XW + W^2$$

$$Y = (kX)^2 + 5(kX)(kW) + (kW)^2$$

$$New Y = k^2(X^2 + 5XW + W^2) = k^2(Y)$$

It's homogenous of degree 2

Returns to scale

This refers to the proportion by which output changes when all inputs are multiplied by the same multiple. For example, assume that a farmer has been using 2 employees and 1 tractor and the resultant output produced is 1000bags of maize.

If the farmer doubles the inputs/factors such that he employs 4 employees and 2 tractors, it can result in three scenarios:

- i) Using 4 employees and 2 tractor production is 2000 bags
This is a case of constant returns to scale. When inputs are doubled, output doubles. The output changes by the same multiple by which inputs were multiplied by
- ii) 4 employees 2 tractors produce 1500 bags

The output has increased by less than double i.e. By 1.5 times. This is a case of decreasing returns to scale. The output increases by less than the multiple by which the inputs were multiplied by.

iii) 4 Employees - 2 tractors - 3000 bags

The output has increased by more than double i.e. by three times. This is increasing returns to scale. The output increases by more than the multiple by which the inputs were multiplied by.

A function is said to display:

- i) Constant returns to scale if its degree of homogeneity is of degree 1.
- ii) Decreasing returns to scale if its degree of homogeneity is less than 1.
- iii) Increasing returns to scale if its degree of homogeneity is more than 1.

Example six

Determine the nature of returns to scale for the following function:

$$\begin{aligned} \text{i) } Q &= 80K^{0.2}L^{0.5} \\ Q &= 80(CK)^{0.2}(CL)^{0.5} \\ Q &= C^{0.2+0.5}80K^{0.2}L^{0.5} \\ Q &= C^{0.7}80K^{0.2}L^{0.5} \end{aligned}$$

This function is homogenous of degree 0.7. Hence it displays decreasing returns to scale.

$$\begin{aligned} \text{ii) } Q &= A L^{\frac{1}{2}} K^{\frac{3}{4}} \\ \text{New } Q &= 80(CL)^{0.5}(CK)^{0.75} \\ \text{new } Q &= C^{1.25}AK^{0.75}L^{0.5} \end{aligned}$$

Function is homogenous of degree 1.25. Hence displays increasing returns to scale.

4.9 EULER'S THEOREM

According to this theorem, if a production function displays characteristics of constant returns to scale, the sum of marginal product of labour multiplied by labour units (L) and marginal product of capital multiplied by capital units (K) is equal to the total output.

i.e. if $Q = A K^{\alpha} L^{\beta}$

Such that $\alpha + \beta = 1$

Then: $MP_L \cdot L + MP_K \cdot K = Q - - - - - 1$

From the production function:

$$Q = AK^{\alpha}L^{\beta}$$

$$9K^3 - 15KL^2 - 3L^3 = 3Q$$

$$3(3K^3 - 5KL^2 - L^3) = 3Q$$

$$Q = (3K^3 - 5KL^2 - L^3)$$

$$3Q = 3Q$$

4.10 Comparative statics

This section deals with how the equilibrium value of an endogenous variable changes when there is a change in any of the exogenous variables or parameters.

Market Model:

Consider the simple one commodity market model given as:

$$Q = a - bP \quad (a, b) > 0$$

$$Q = -c + dP \quad (c, d) > 0$$

$$P^* = \frac{a + c}{b + d}$$

$$Q^* = \frac{ad - bc}{b + d}$$

P^* and Q^* are the endogenous variables and they have been purely expressed as a function of the parameters a , b , c , and d .

We would want to analyze how P^* changes due to change in any of the parameters. This is done by partially differentiating P^* with respect to any of the parameters, by making use of the quotient rule: e.g

$$\frac{\partial P^*}{\partial a} = \frac{1}{b + d} > 0 \quad \frac{\partial P^*}{\partial c} = \frac{1}{b + d} > 0$$

$$\frac{\partial P^*}{\partial b} = \frac{-1(a + c)}{(b + d)^2} < 0 \quad \frac{\partial P^*}{\partial d} = \frac{-1(a + c)}{(b + d)^2} < 0$$

Since a , b , c and d are greater than 0, the expressions $\frac{\partial P^*}{\partial a}$ and $\frac{\partial P^*}{\partial c} = \frac{1}{b + d} > 0$

And

$$\frac{\partial P^*}{\partial b} \text{ and } \frac{\partial P^*}{\partial d} = \frac{-1(a + c)}{(b + d)^2} < 0$$

How does the equilibrium Q^* change as parameters change. This is done by getting the partial derivatives of Q^* with respect to a , b , c and d .

$$\frac{\partial Q^*}{\partial a} = \frac{d}{b+d} > 0 \quad \frac{\partial Q^*}{\partial c} = \frac{-b}{b+d} < 0$$

$$\frac{\partial Q^*}{\partial b} = \frac{-d(a+c)}{(b+d)^2} < 0 \quad \frac{\partial Q^*}{\partial d} = \frac{b(a+c)}{(b+d)^2} > 0$$

National income model

Simple national income model:

$$Y = C + I_0 + G_0$$

$$C = c_0 + c_1(Y^d) \text{ where } c_0 > 0 \quad 0 < c_1 < 1$$

$$T = t_0 + t_1 Y \quad t_0 > 0 \quad 0 < t_1 < 1$$

$$Y^d = Y - T$$

$$Y = c_0 + c_1(Y^d) + I_0 + G_0$$

$$Y = c_0 + c_1(Y - T) + I_0 + G_0$$

$$Y = c_0 + c_1(Y - t_0 - t_1 Y) + I_0 + G_0$$

$$Y = c_0 + c_1 Y - c_1 t_0 - c_1 t_1 Y + I_0 + G_0$$

$$Y - c_1 Y + c_1 t_1 Y = c_0 - c_1 t_0 + I_0 + G_0$$

$$Y(1 - c_1 + c_1 t_1) = c_0 - c_1 t_0 + I_0 + G_0$$

$$Y^* = \frac{c_0 - c_1 t_0 + I_0 + G_0}{(1 - c_1 + c_1 t_1)}$$

Y^* is the equilibrium value of national income.

How does Y^* change as exogenous variables change.

$$\frac{\partial Y^*}{\partial I_0} = \frac{1}{(1 - c_1 + c_1 t_1)} \text{ This is the autonomous investments multiplier.}$$

Given that

c_1 and $c_1 t_1$ are positive and less than 1, the autonomous investment multiplier is positive, meaning an increase in the autonomous investments will lead to an increase in the equilibrium value of the national income.

Using partial differentiation, we can also get the autonomous tax multiplier.

$$\frac{\partial Y^*}{\partial I_0} = \frac{-c_1}{(1 - c_1 + c_1 t_1)}$$

The autonomous tax multiplier is negative. This means that an increase in taxes will lead to a decrease in the national income.

Similarly, we can get the tax rate multiplier. It tells us how the national income changes due to a unit change in the tax rate.

$$\begin{aligned}\frac{\partial Y^*}{\partial t_1} &= \frac{-(c_1)(c_0 - c_1 t_0 + I_0 + G_0)}{(1 - c_1 + c_1 t_1)^2} \\ &= \frac{-(c_1)Y^*}{(1 - c_1 + c_1 t_1)}\end{aligned}$$

An increase in the tax rate reduces the value of the national income and vice versa.

National income model for an open economy

$$Y = C + I_0 + G_0 + X_0 - M$$

$$C = c_0 + c_1(Y^d) \text{ where } c_0 > 0 \quad 0 < c_1 < 1$$

$$T = t_0 + t_1 Y \quad t_0 > 0 \quad 0 < t_1 < 1$$

$$Y^d = Y - T$$

$$M = m_0 + m_1 Y$$

$$Y = c_0 + c_1(Y^d) + I_0 + G_0 + X_0 - (m_0 + m_1 Y)$$

$$Y = c_0 + c_1(Y - T) + I_0 + G_0 - X_0 - m_0 - m_1 Y$$

$$Y = c_0 + c_1(Y - t_0 - t_1 Y) + I_0 + G_0 - X_0 - m_0 - m_1 Y$$

$$Y = c_0 + c_1 Y - c_1 t_0 - c_1 t_1 Y + I_0 + G_0 - X_0 - m_0 - m_1 Y$$

$$Y - c_1 Y + c_1 t_1 Y + m_1 Y = c_0 - c_1 t_0 + I_0 + G_0 - X_0 - m_0$$

$$Y(1 - c_1 + c_1 t_1 + m_1) = c_0 - c_1 t_0 + I_0 + G_0 - m_0$$

$$Y^* = \frac{c_0 - c_1 t_0 + I_0 + G_0 - m_0}{(1 - c_1 + c_1 t_1 + m_1)}$$

We can get a number of multipliers.

$$\frac{\partial Y^*}{\partial G_0} = \frac{1}{(1 - c_1 + c_1 t_1 + m_1)}$$

This is the government multiplier. An increase in government expenditure causes the national income to increase.

$$\frac{\partial Y^*}{\partial I_0} = \frac{1}{(1 - c_1 + c_1 t_1 + m_1)}$$

$$\frac{\partial Y^*}{\partial M_0} = \frac{-m_0}{(1 - c_1 + c_1 t_1 + m_1)}$$

Get:

$$\frac{\partial Y^*}{\partial c_1}, \frac{\partial Y^*}{\partial m_1} \text{ and } \frac{\partial Y^*}{\partial t_1}$$

ASSIGN

1. A total cost function is given by:

$$C = Q_2^3 + Q_1 Q_2 - 6Q_2^3 + 40$$

Find: MC_1 , MC_2 and if function displays increasing or decreasing MC with respect to each of the goods.

2. Find the degree of homogeneity and nature of returns to scale for the following function

$$Q = 10 K^{1/4} L^{3/4}$$